

Robotics in Syria Report

December 2017



DISCLAIMER

This report highlights the experiences and needs of Syrian Robotics community in the country and shows other events, competitions, and activities of other countries around the world in goal to support robotic culture in this country. This report shows how we can implement or simulate this solution inside the Syrian community. This report shows a very close view of the author. The study aims to present policy and programme recommendations for use by governments, international NGOs, donors and other educational institutions, with the ultimate goal of helping youths get access to these kinds of sciences and activities.

The analysis, results, and recommendations in this report represent the opinions of the author and are not necessarily representative of the position of the reviewers.

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**“THE BEST
WAY TO
PREDICT THE
FUTURE IS TO
INVENT IT”**

-ALAN KAY



This isn't a robot

This isn't a robot

It very well may have hydraulic arms a drivetrain and a circuit board but this isn't a robot

This is a lesson in humility hard work and collaboration

This is a CrossFit workout for the muscle upstairs

This is an apple for the next generation of Newtons

It's a lot of things but this isn't a robot

This is a road map that leads straight to engineering

this is a rock to shatter a thousand and glass ceilings

A rope to pull a generation up and out of poverty a

sledgehammer to break down

cultural barriers

A universal language spoken in every nation on earth

and because of that this is a compass

That can help point the way to peace but please don't think this is just a robot

This is Glastonbury for technology

This is Mission Control for the next moonshot

This is the rosetta stone to help translate the future

This it the sport that will help us win tomorrow

This is the software to program ourselves to cure cancer build a clean engine or step foot on Mars but this isn't a robot

It's a machine to build the people who will change the world you

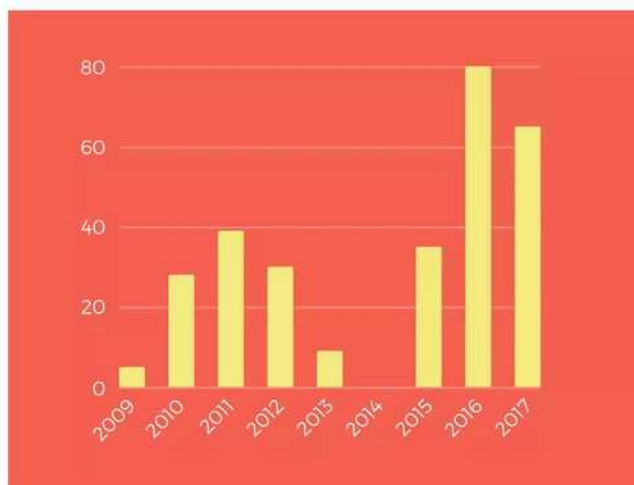
Statistics

Teams Number

From the beginning of Robotics in Syria in 2009 until 2017 there were almost 291 teams in different competitions and in different categories. As distributed by date:

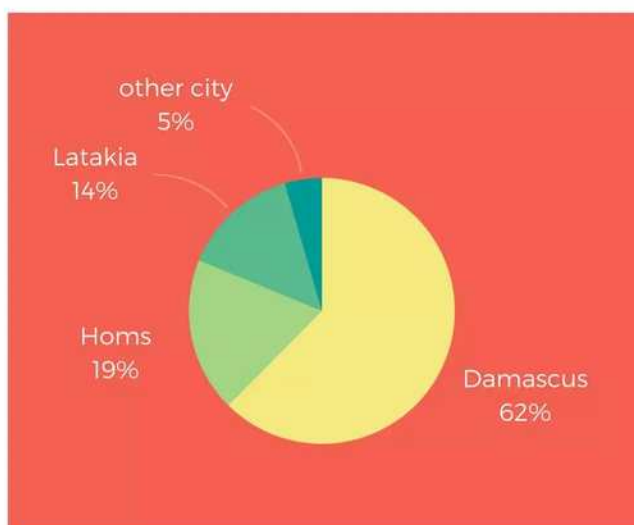
Distribution by date

- 2009: 5 Teams
- 2010: 28 Teams
- 2011: 39 Teams
- 2012: 30 Teams
- 2013: 9 Teams
- 2014: 0 Teams
- 2015: 35 Teams
- 2016: 80 Teams
- 2017: 65 Teams



Geographical Distribution

- Damascus: 239
- Homs: 25
- Hama: 2
- Tartous: 2
- Latakia: 19
- Sweida: 2
- Aleppo: 2



Statistics

Robotic Club Number

24

is the number of Robotics Club
inside Syria.

5

Robotics Club outside Damascus.

What is Robotic Club?

We mean by mentioning a robotic club is kind a place that can provide the tools and the coaches for the students and train the students themselves so it's kind of school for robotics. And here is a list of all robotics clubs that work in 2016 and 2017.

Note: Since 2009 until 2015 many teams have been participating independently or through sponsors indirectly.

- Damascus University Faculty of Mechanical and Electrical Engineering
- Damascus University Faculty of Information Engineering
- University of Kalamoon
- Yarmouk Private University
- International University for Science and Technology
- Syrian International University for Science and Technology
- Arab International University
- DTC-UNRWA
- Higher Institute for Applied Sciences and Technology
- National Center for the Distinguished
- Syrian Computer Society - Damascus, Homs, Hama, Sweida, Aleppo, Tartous.
- Dar Al Aman
- Dar Al Rahma
- Lahn Al-Heat
- Hadara Center
- Qiam Center
- AL-Rabta Organization
- Sawa School
- Al-Wataniah Syrian National School



Syrian teams in OARC 2011 Lebanon

HISTORY OF ROBOTICS IN SYRIA

2009

- The first ever robotic competition occurring in Syria and that was a very small launch for this kind of event and it started with a FLL (FIRST LEGO League) Mars mission competition.

2010

- The first official event for FLL Body Forward Challenge, this was the first event in Syria that was organized by a private sector company "Idea" that succeeds to get a lot of sponsors. This event was held in AL-Sharq Club in Damascus with about 28 team 2 of these teams came from Homs city.

2011

- FLL Food Factor Challenge this was the last FLL competition happening in Syria because of the FLL or as FIRST LEGO League was a nonprofit organization registered in the USA, so by the law of the US Sanctions and Foreign Trade Department, which prohibits dealing with Syria. There was almost 39 team and only 4 team came from Homs city.
- This year was the first year ever Syria joining the first regional competition outside the country and that was OARC (Open Arabic Robotic Competition) in 2011 which was hosted in Lebanese American University, Byblos, Lebanon. With more 100 team from 12 Arab country in MENA.

2012

- In this year was the first launch for WRO (World Robot Olympiad) in Syria, and hosting this event in 2012 was kind of challenging because of the crisis and the beginning of the war that Syria was living at the time, there was only 30 teams from Damascus and this year was the first where none from outside Damascus and other cities joining the competition.
- Was the first ever Global participation for Syrian teams in the finals for WRO that was hosted in UAE with only with one team.

HISTORY OF ROBOTICS IN SYRIA

2013

- In this year the situation in Syria deteriorated further, however, despite the circumstances, the competition still launched, but in a very small number of teams with a lot of buzz for the event.

2015

- The third event of WRO was launched and happened in Damascus University hall with 35 teams and this event was the first co-organized one by Idea company and SCS organization (Syria Computer Society).
- Was second participation for Syria team in a global competition that was hosted in Russia with 2 teams, and that was the first participation for 2 judges from Syria.

2016

- The fourth WRO event on Syria was hosted completely by SCS after SCS got official license for WRO event in Syria, there were 80 teams from Damascus and almost 6 teams from different cities in Syria like Homs, Hama, Taurus, and Sweida.
- The third global participation in WRO India with 8 teams in all categories of the competition with 1 judge for this event. Syria got 2 cups in this competition with 2 teams in best 15 team.
- Was the first National Sumo Robot Championship in Syria with almost 15 teams from Damascus, Latakia and Homs city with small competition for "Lego Space Challenge".
- Was the first time Syria participated in EDC (Engineering Design Challenge) that was hosted and organized by AUB (American University of Beirut) Robotics Club with one team. Syria got 1 cup in this competition.
- Was the first time Syria participated on ARC (Annual Robotic Competition) in Lebanon as regional competition with 1 team. Syria got 1 cup in this competition too.

HISTORY OF ROBOTICS IN SYRIA

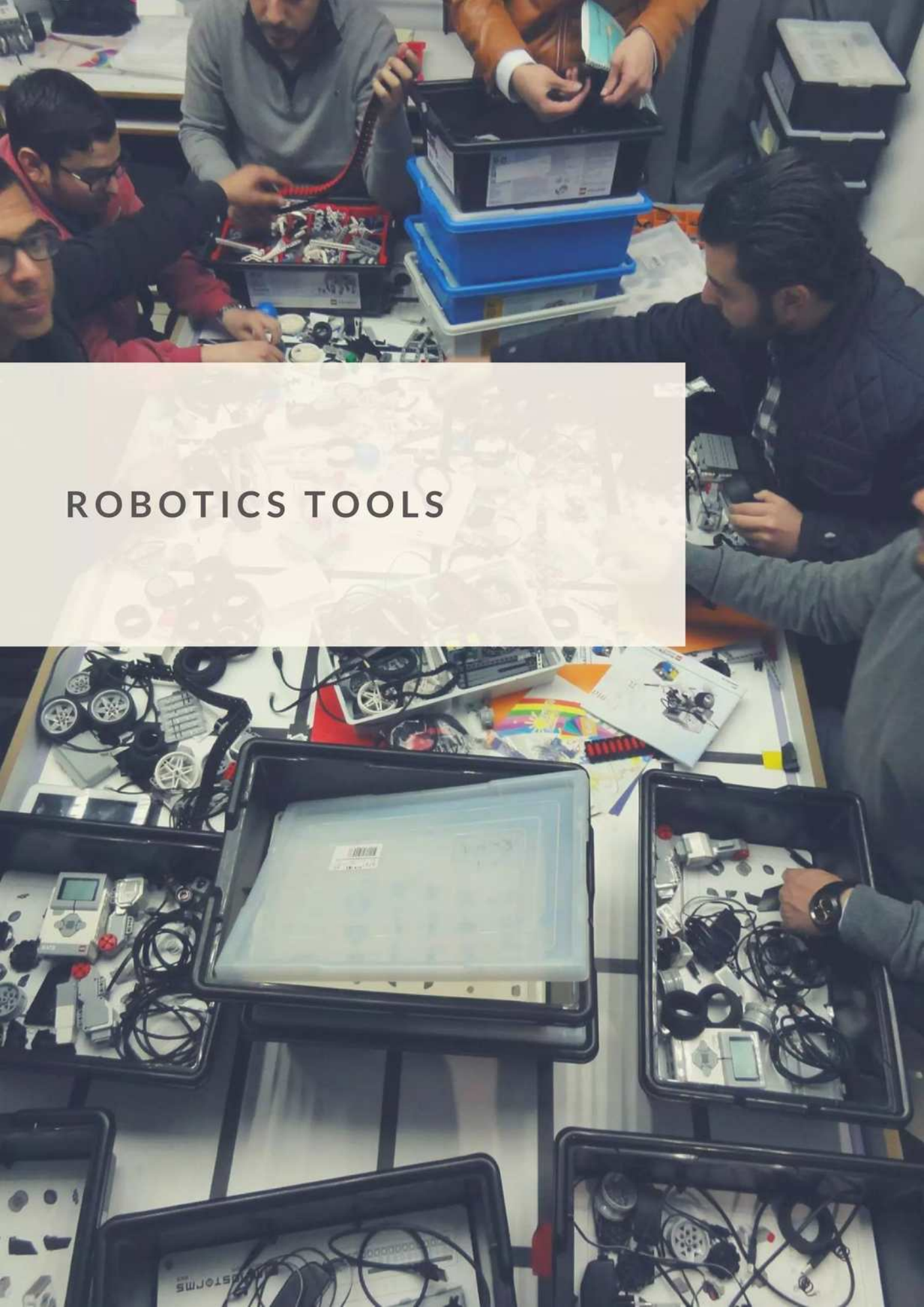
2017

- Was the first launch for ARC in Syria, this local competition was hosted and organized by SCS.
- The winner's team for the local event in ARC participation in ARC competition in Beirut, Lebanon with Partnership with Kalimat Association who hosted and organized the ARC competition in Lebanon with almost 400 teams from different private and public school in Lebanon. And there were 5 Syrian teams participating in Lebanon's, Syrian teams got 2 cups in this ARC 3 on Lebanon.
- A small local competition for EDC in Syria was launched, hosted with some individual team.
- Syria got officially the hosting and organizing rights for SCS for FGC (FIRST Global Challenge) but Syrian teams could not participate or even get the tools for the competition because this competition was hosted in USA.
- The 5 edition of WRO event in Syria, with 70 teams from around Syria.
- The fourth global participation for WRO that was hosted in Costa Rica with 6 teams and 2 judges, Syria gott 1 cup in this competition with 1 team in best 5 teams from around the world.
- Syria will participate with 1 team for the first time in IRC (International Robotics Challenge) India as part of Techfest, which is the annual science and technology festival of Indian Institute of Technology Bombay.



First WRO event in Syria 2011

In Arab Cultural Center in Adawi - Damascus



ROBOTICS TOOLS

1- LEGO



LEGO MINDSTORMS

Lego Mindstorms is a hardware-software platform produced by Lego for the development of programmable robots based on Lego building blocks. Each version of the system includes an intelligent brick computer that controls the system, a set of modular sensors and motors, and Lego parts from the Technic line to create the mechanical systems.

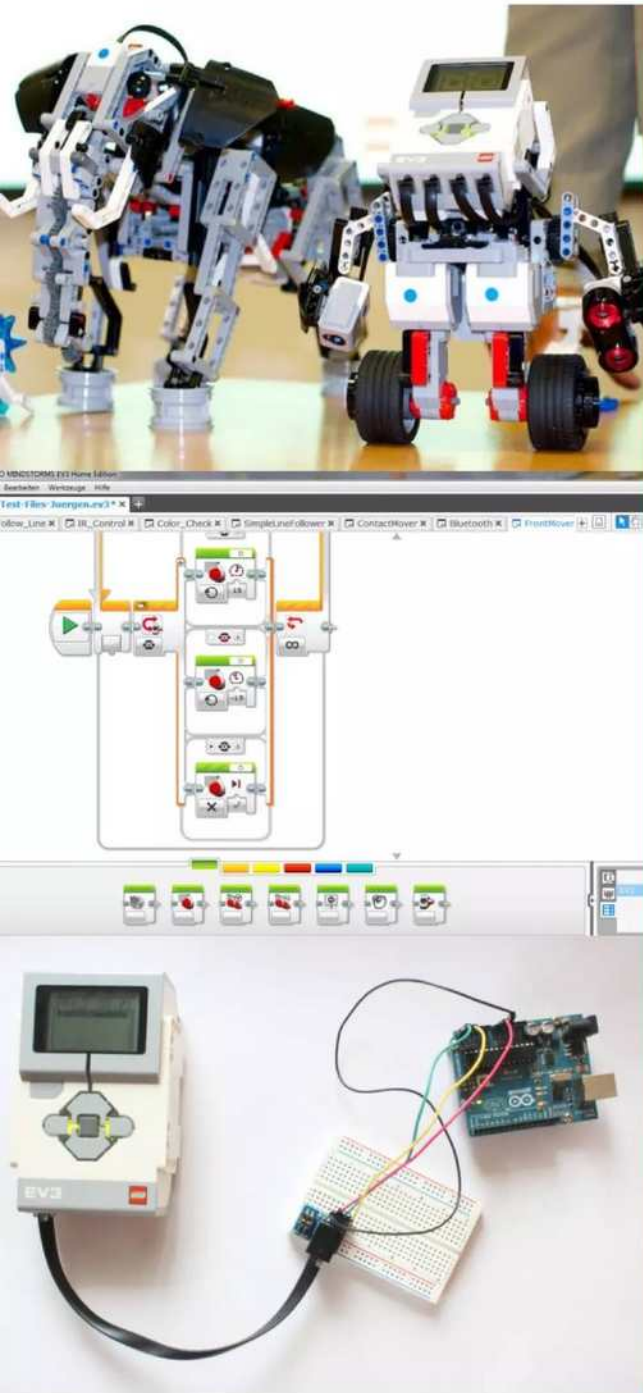
Since creation, there have been 4 generations of the Mindstorms platform: the original Robotics Invention System, NXT, NXT 2.0, and EV3. With each platform release, the motor and sensor capabilities expanded. The latest system, Lego Mindstorms EV3, was released on September 1, 2013. Some robot competitions use this set, such as the FLL (First Lego League) and the WRO (World Robot Olympiad).

PROS:

- It has a long lifespan of work; as it can be reused for many seasons thanks for its high quality production.
- It has a large community base that provides support, as in building instructions and online help forum.
- It's favorable by kids and usable for them as one of LEGO part.
- They have one of the greatest and easiest User Interface for programming in a very fun and easy way for kids, with the ability for higher programing level for for older ages and more professional users.
- It supports so many add-in and a lot of accessories that can get the sensors and controller to next level.
- They have a special set for kids under 9 years old, called WeDo 2.0 it has a motor and very super easy control with great user interface for coding that supports apps for tablets like iPad and other android tablets and mobile phones which make coding for kids more fun and easy for them. !There a pilot lunch for this age of catorger in WRO.

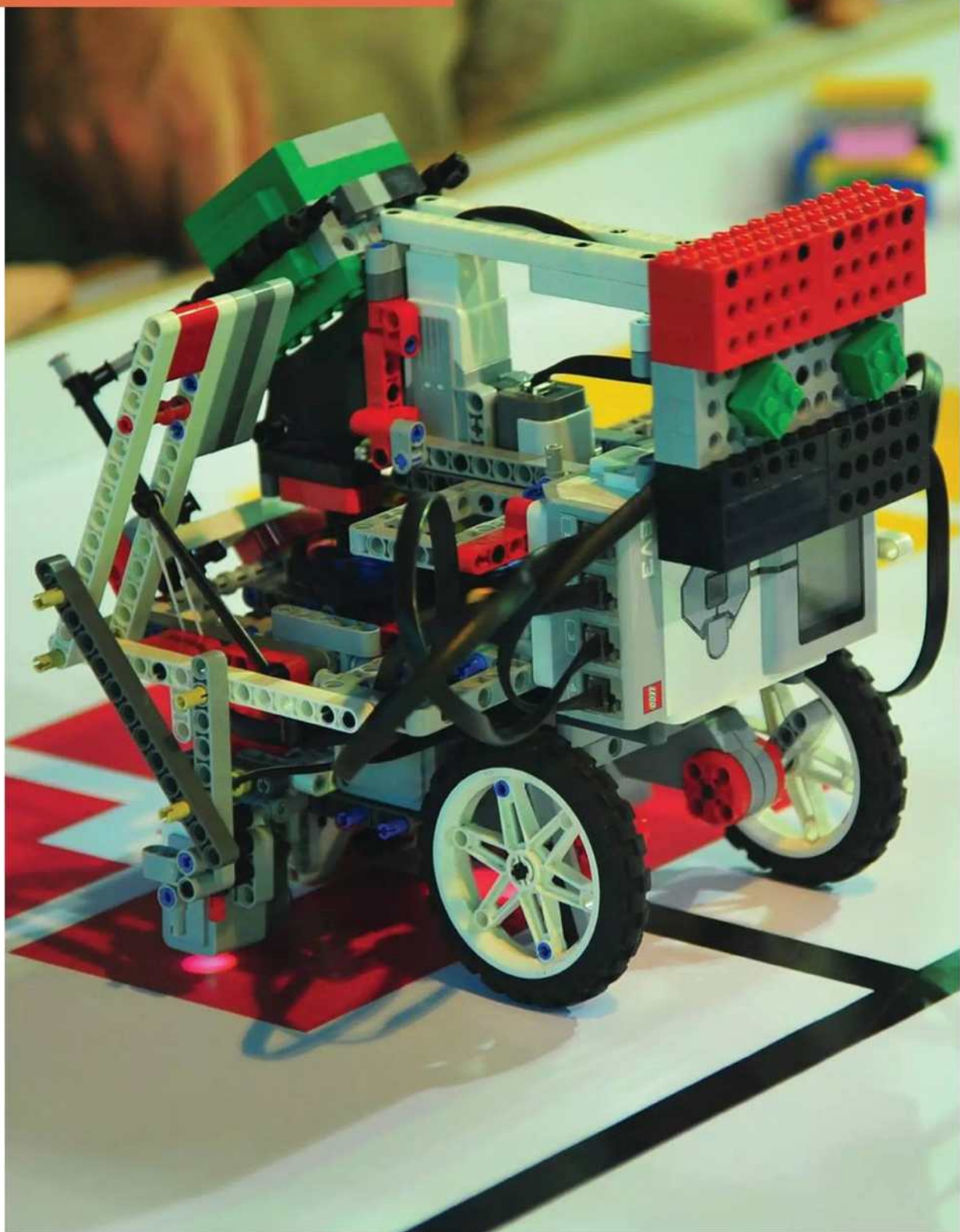


CONS:



- It is super expensive, to get a whole set for proper one team to WRO competitions as expsalema it costs (1 LEGO Core, 1 LEGO Expectations part, 2 Hi-tech color sensors, 1 exstar motor, 1 Charger) about 900\$ for one team only.
- It's so hard to repair or maintain.
- You have a big limitations of type for parts you can use it in building Robots for other competitions or non-lego project.

WRO SYRIA 2016





2- TETRIX

TETRIX ROBOTICS BY PITSCO

TETRIX Robotics consists of two robotic kits by Pitsco, Company. intended for educational purposes and competitions such as the FTC (FIRST Tech Challenge.), WRO (World Robot Optimality) and SkillsUSA, Urban Search & Rescue (USAR). The sets are TETRIX MAX for high school and TETRIX PRIME for middle school.

They are two main type of TETRIX Collocation and every one of them have their own purpose for STEM education tools or to join the competition.

TETRIX PRIME:

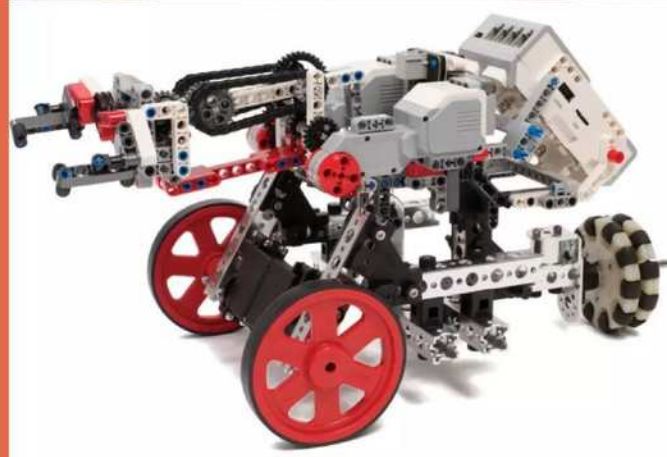
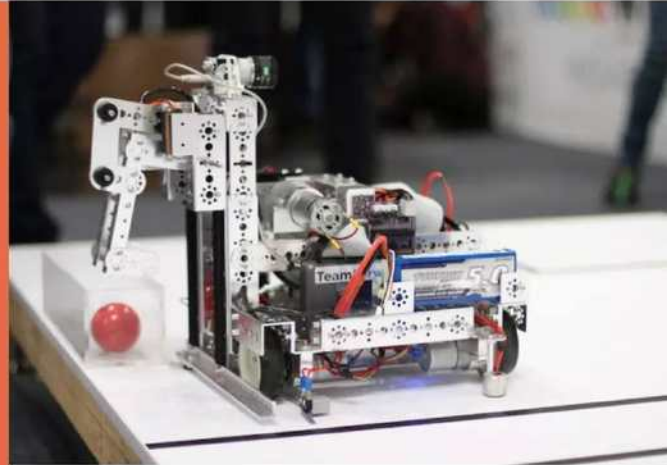
- Top Benefit: Affordable
- Sets can be used year after year with teams of up to four students, and numerous builds can be constructed with just the PRIME Starter Set.
- Age appropriate for students ages 12 and older.
- Easy building and disassembling of a prototype (30-45 minutes).
- Excellent for engineering, performs multiple design iterations.
- Compatible with TETRIX MAX elements – congruent hole patterns and fasteners ensure true integration of elements.
- Control Elements: TETRIX PULSE Robotics Controller, LEGO MINDSTORMS EV3 or NXT Brick, Arduino UNO, myRIO

TETRIX MAX:

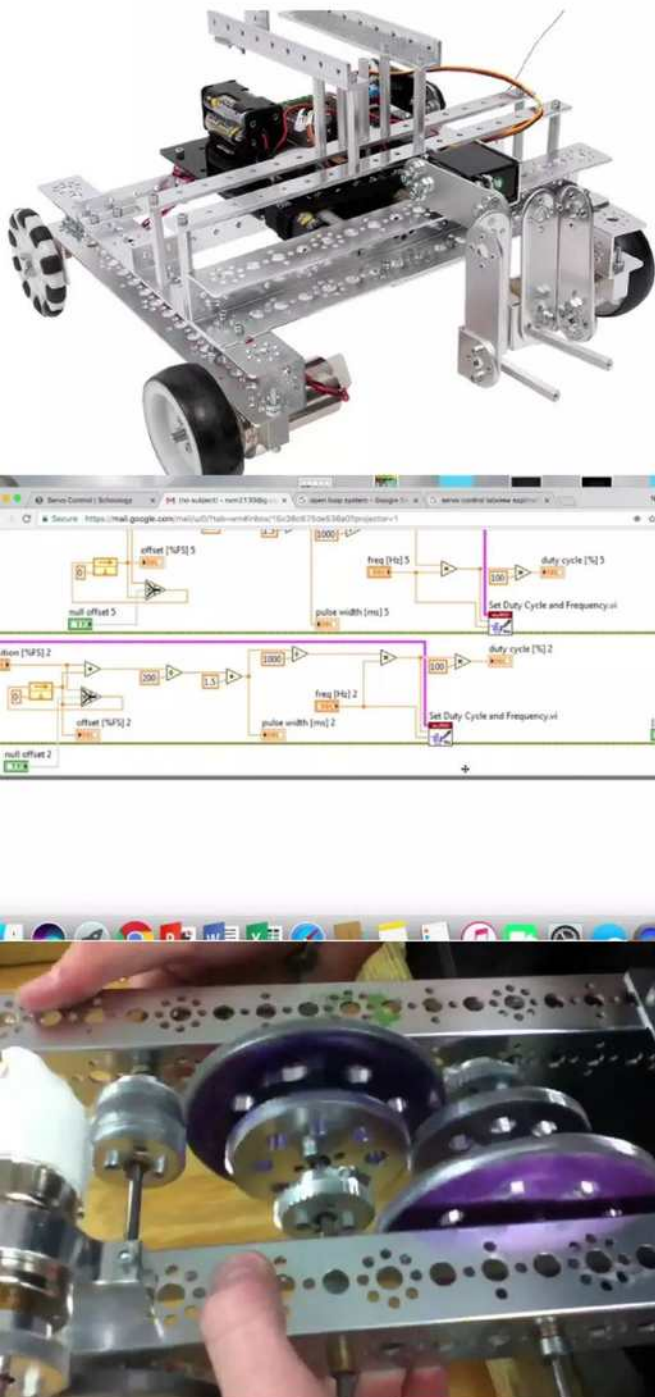
- Top Benefit: Comprehensive
- More than 100 elements to choose from, and several builds can be constructed with just the MAX Starter Set.
- Age appropriate for students ages 14 and older.
- Ideal for student competitions – durable for large-scale robots and provides secure connections.
- Robot building system of choice for the FIRST® Tech Challenge.
- Compatible with TETRIX PRIME elements – congruent hole patterns and fasteners ensure true integration of elements.
- Control Elements: TETRIX PULSE Robotics Controller, LEGO MINDSTORMS EV3 or NXT Brick

PROS:

- It is considered as next level for LEGO kit, thanks for its metal material parts. It can make the robot sturdier and do new tasks without the limitations of plastic parts like other robots.
- They have very strong motors, that make robots move and torque, it takes speed for robots using it to another level.
- You have many options to control this motor and it can support several types of sensors from other kits and sets, it can support LEGO EV3, NXT, myRio, Arduino controller.
- You can use many options for coding and programming the robot like Robotic C, LabVlew, LEGO Mindstorm and more.
- They have a lot and big extensions and metal gear sets that really make very complex robots and are easy for students to use.



CONS:



- It's considered very expensive with higher costs than the LEGO.
- It can not work without a controller bought separately from the set.
- The set does not contain any sensors, all sensors must be obtained from other sets completely like LEGO Mindstorm kit sensors.
- One set or kit is not enough for the student to join one of the competitions, most of robot tasks require more parts and motors that should be bought and obtained separately.
- It's only available in USA, and it's so hard to ship or get sets and extra extensions, especially for Syria because of economic sanctions on the country of US products.



FTC (FIRST TECH CHALLENGE)



3- MATRIX



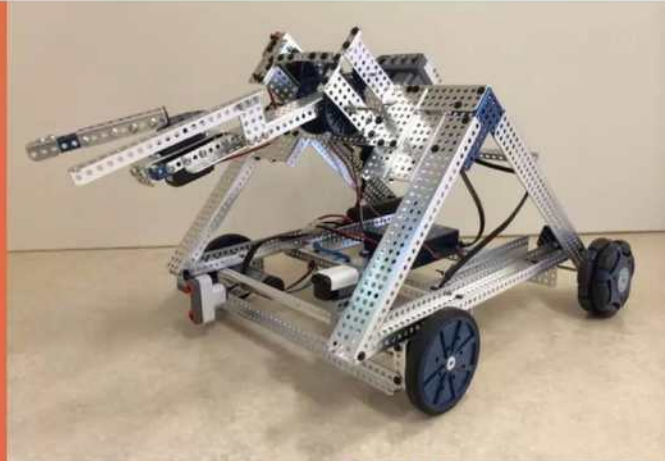
MATRIX ROBOTICS KIT

The Matrix Robotics Kit of Parts is an aluminum robotics building system that is fully compatible with the LEGO Mindstorms system. The kit has various aluminum frames to build a more robust robot, more professional engines (DC motors with encoders), able to move the robot precisely on difficult terrain, large wheels ideal for odometry, Omni Wheels.

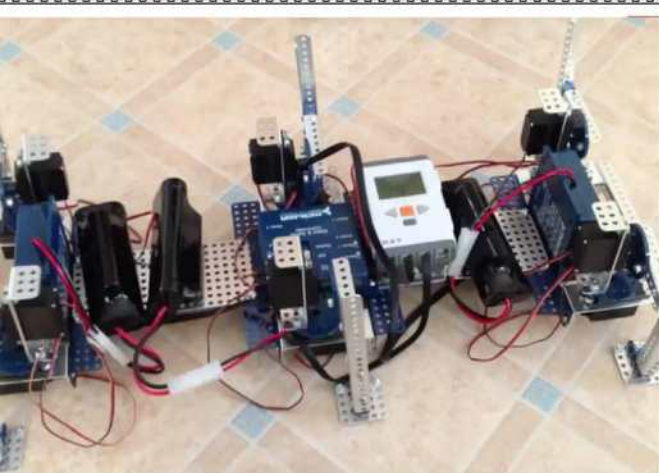
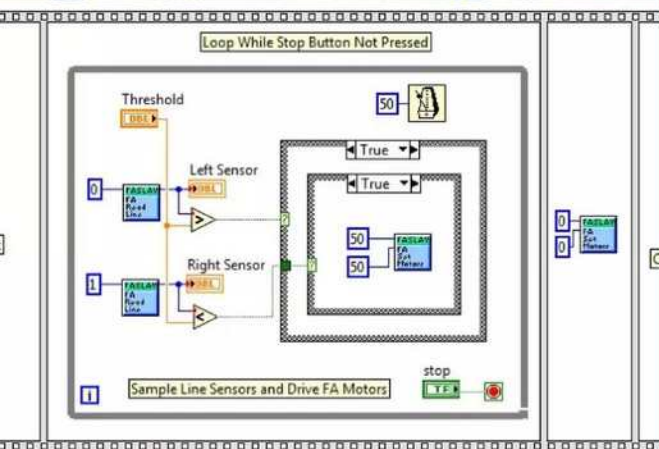
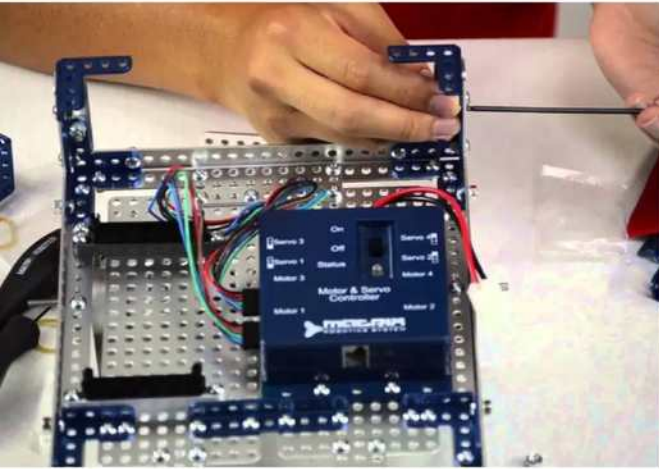
The kit supports the FTC (FIRST Tech Challenge.), WRO (World Robot Olympiad) only. And you must own Matrix controller write LEGO Mindstorm NXT or EV3 to control this Motor inside the kit. The kit does not contain sensors as well.

PROS:

- It is considered as next level for LEGO kit, thanks for its metal material parts. It can make the robot sturdier and do new tasks without the limitation of plastic parts like other robots.
- They have their own set of sensors and tools and those sensors are more advanced than many of the basic sensors that come from LEGO MindStorm kit.
- They have very strong motors, that make the robots move and torque, it takes speed for robots using it to another level.
- They have a lot and big extensions and metal gear sets that really make very complex robots and are easy for students to use.
- You can use many options for coding and programming the robots like Robotic C, LabVlew, LEGO Mindstorm and more.

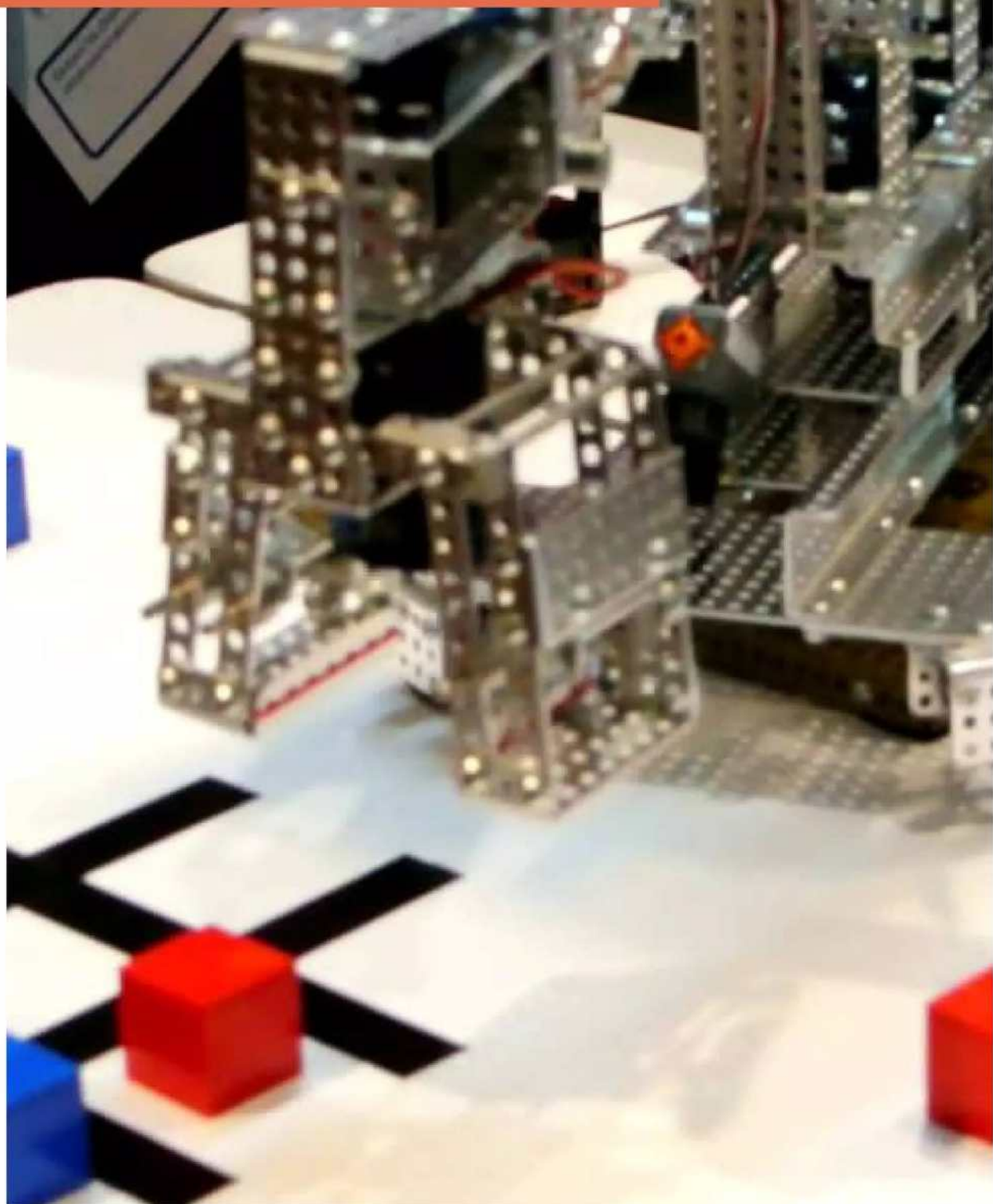


CONS:



- It's considered very expensive with higher cost than the LEGO, and TETRIX.
- It can not work without a controller which is bought separately from the set.
- The set does not contain any sensors, all sensors must be obtained from other sets completely like LEGO Mindstorm kit sensors.
- One set or kit is not enough for the student to join one of the competitions, most of robots task require more parts and motors that should be bought and obtained separately.
- It's only available in USA, and it's so hard to ship or get sets and extra extensions, especially for Syria because of economic sanctions on the country of US products.

MATRIX MECHANICAL ARM IN WRO 2012

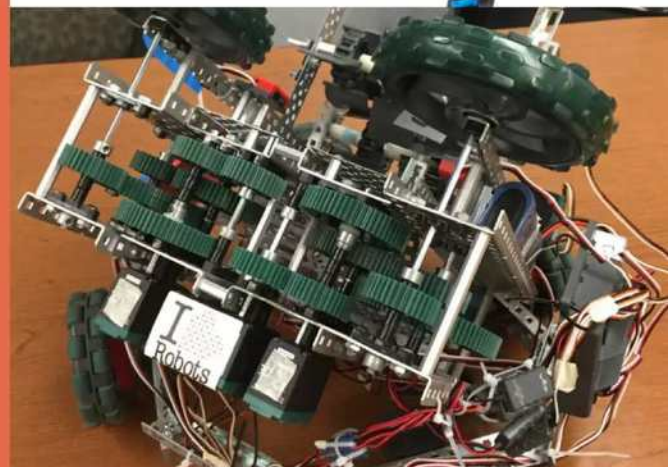


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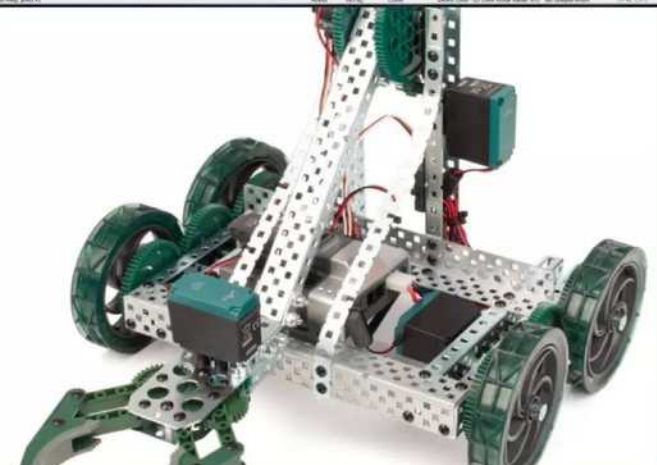
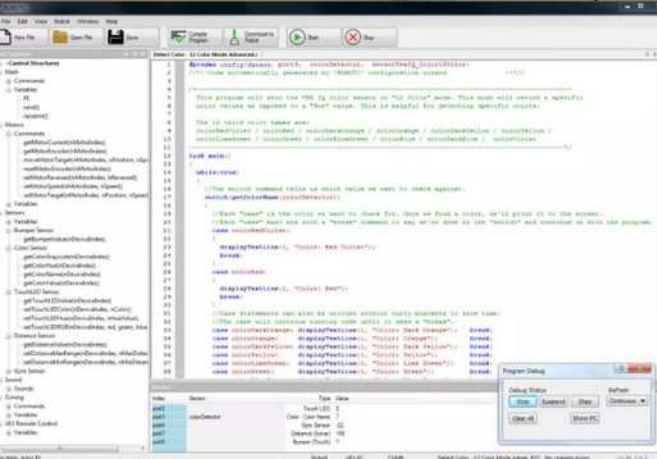
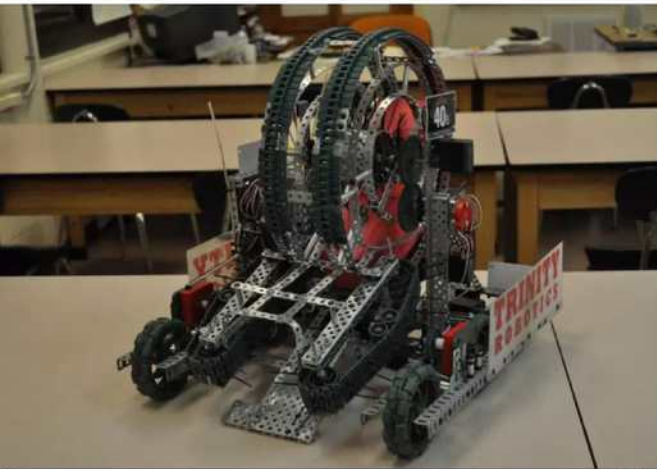
VEX IQ kit Solutions for Elementary & Middle School. Snap-together parts, approachable programming, and modular curriculum make VEX IQ the best for kids in Elementary & Middle School. There is another version of VEX kit named VED EDR Solutions for Middle / High School and Beyond.

PROS:

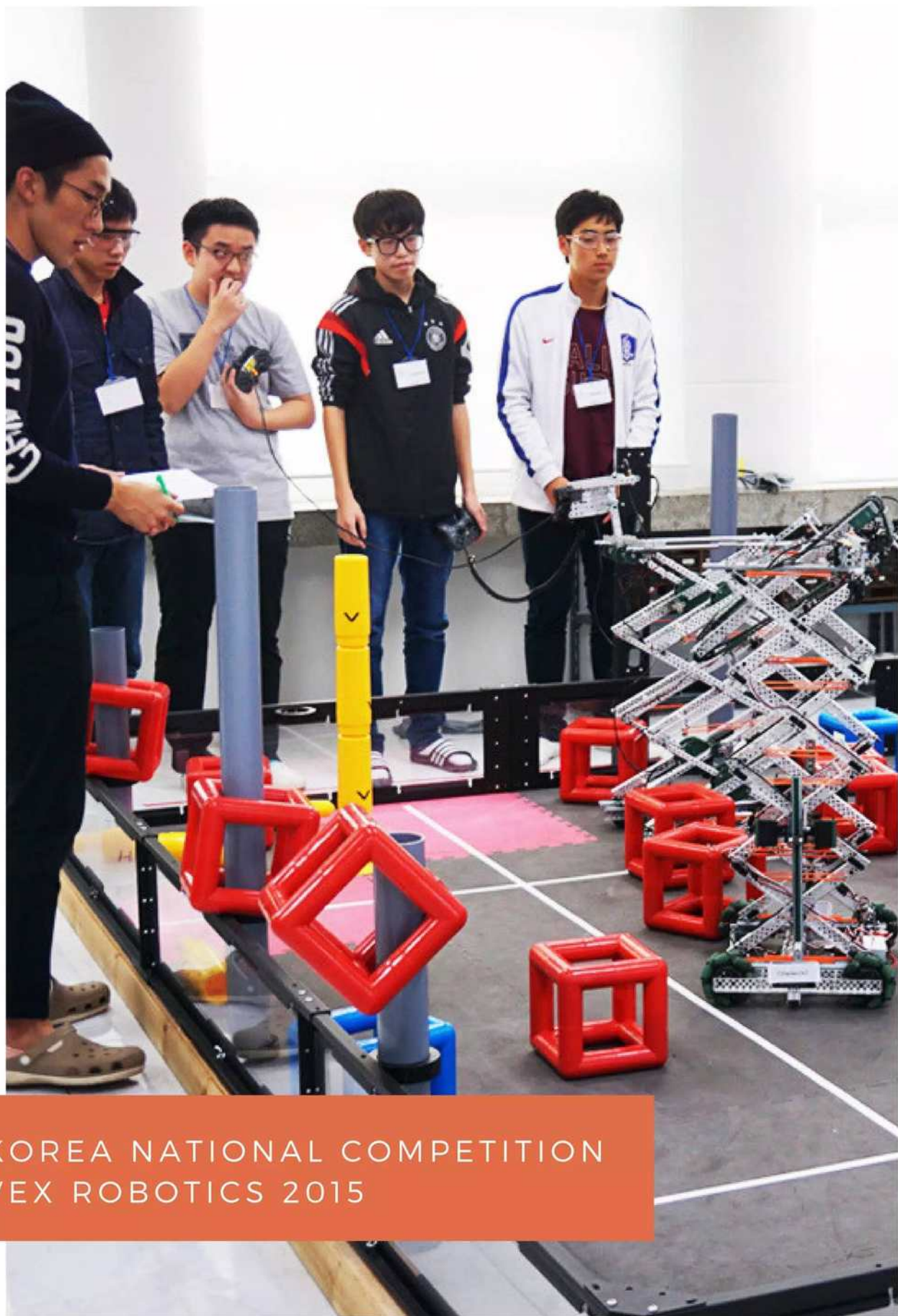
- They have two types of kits: VEX IQ one for elementary and middle school, this comes with plastic parts which make it good for young kids, the other type is: VEX EDR for high school comes with mattel parts.
- it's the cheapest kit in cost in comparison with LEGO Mindstorm, TETRIX and Matrix.
- the kit comes with its own sensors and own controller.
- they have new a excited community that started to show in the last 2 years around the world and the region.



CONS:

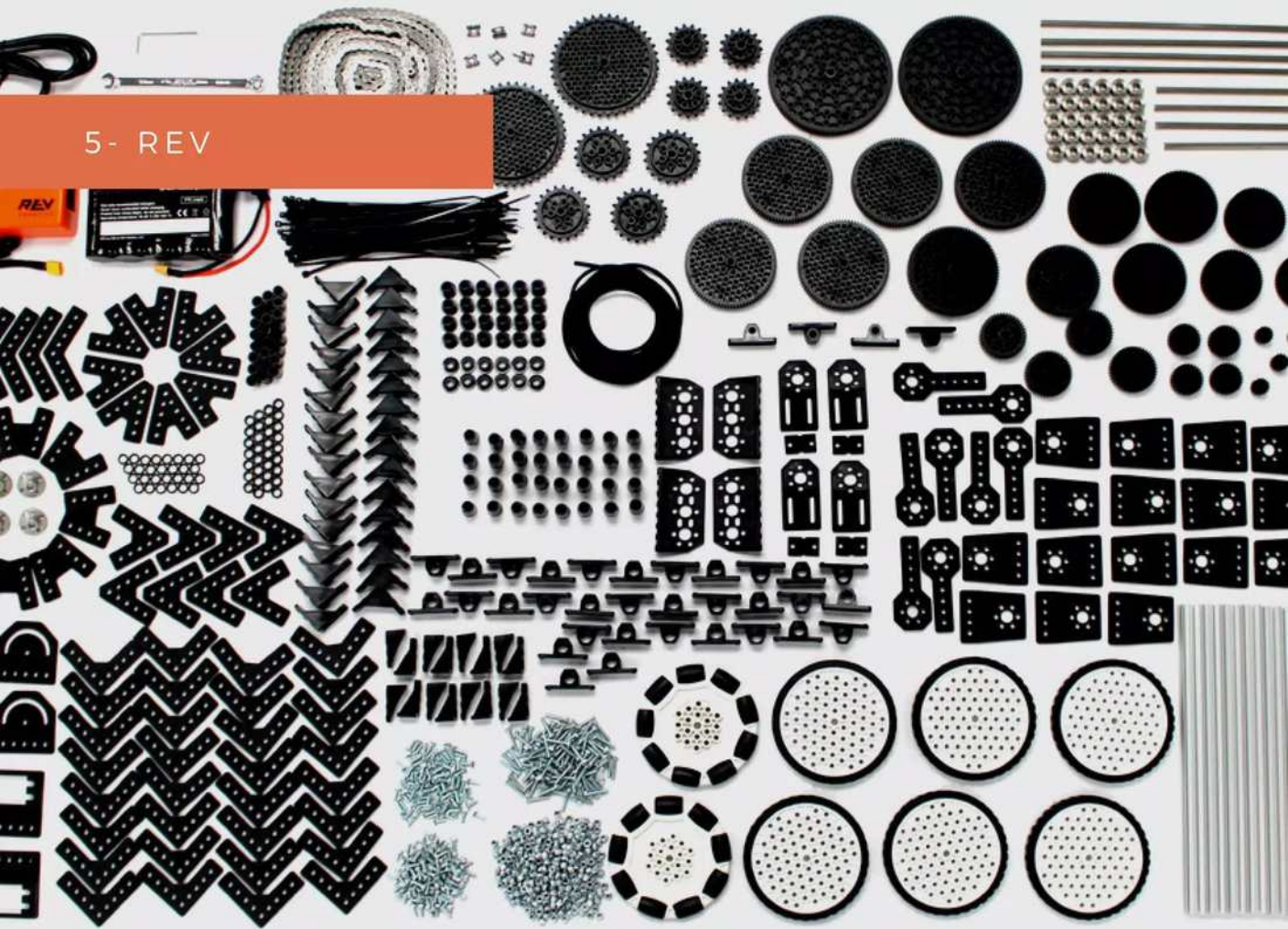


- It does not support any other controller except their own VEX controller, and that goes the same for the sensors.
- There is just one competition that supports VEX so far, however, recently, it's getting more support for VEX robot in FIRST competition.
- They do not have a big community and support around the world like LEGO Mindstorm or TETRIX.
- One set or kit is not enough for the student to join one of the competitions, most of robots tasks require more parts and motors that should be bought and obtained separately.
- There are not many options for coding VEX.
- It's only available in USA, and it's so hard to ship or get sets and extra extensions, especially for Syria because of economic sanctions on the country of US products.



KOREA NATIONAL COMPETITION
VEX ROBOTICS 2015

5- REV



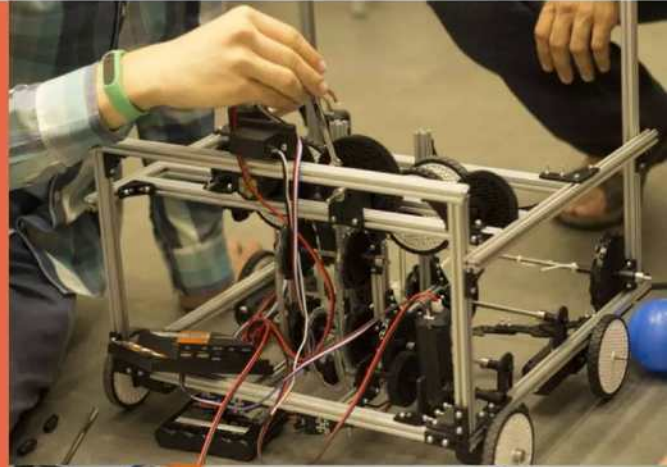
REV ROBOTICS

It is a new company that produces a full Robotics kit with their own motors, sensors and own controller depending on Java programming language. This kit can only be used in FTC, FCR and First Global Competition. They are produced as the company sees that their kits are designed by FIRST mentors made for FIRST students creating FIRST champions.

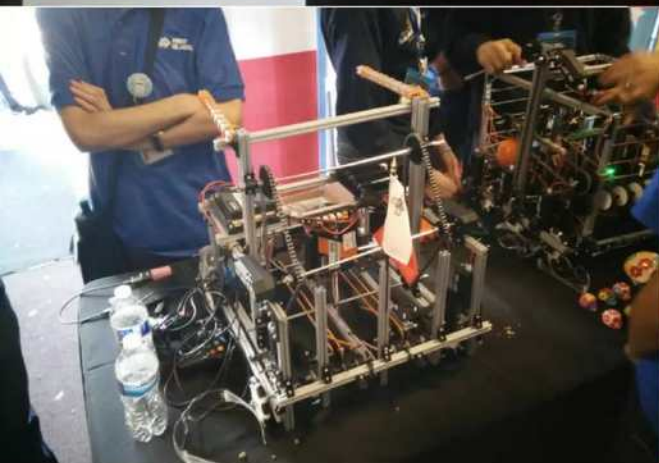
It's a new kit in the market with a small community so far, but with full support from the FLL (First Lego League) it's a very popular option for students and schools and even Robotics Club sto have this kit, because all the tests and materials and resources for the competitions will be supported as the organizers put this kit as first on their mind.

PROS:

- The kit comes with its own sensors, motor and own controller.
- They have new a excited community that started to show in the last 2 years around the world and the region. Thanks for FLL community.
- They have average prices in comparison with other Robotics kits.
- You have the option to buy any extra part separately in an easy way.

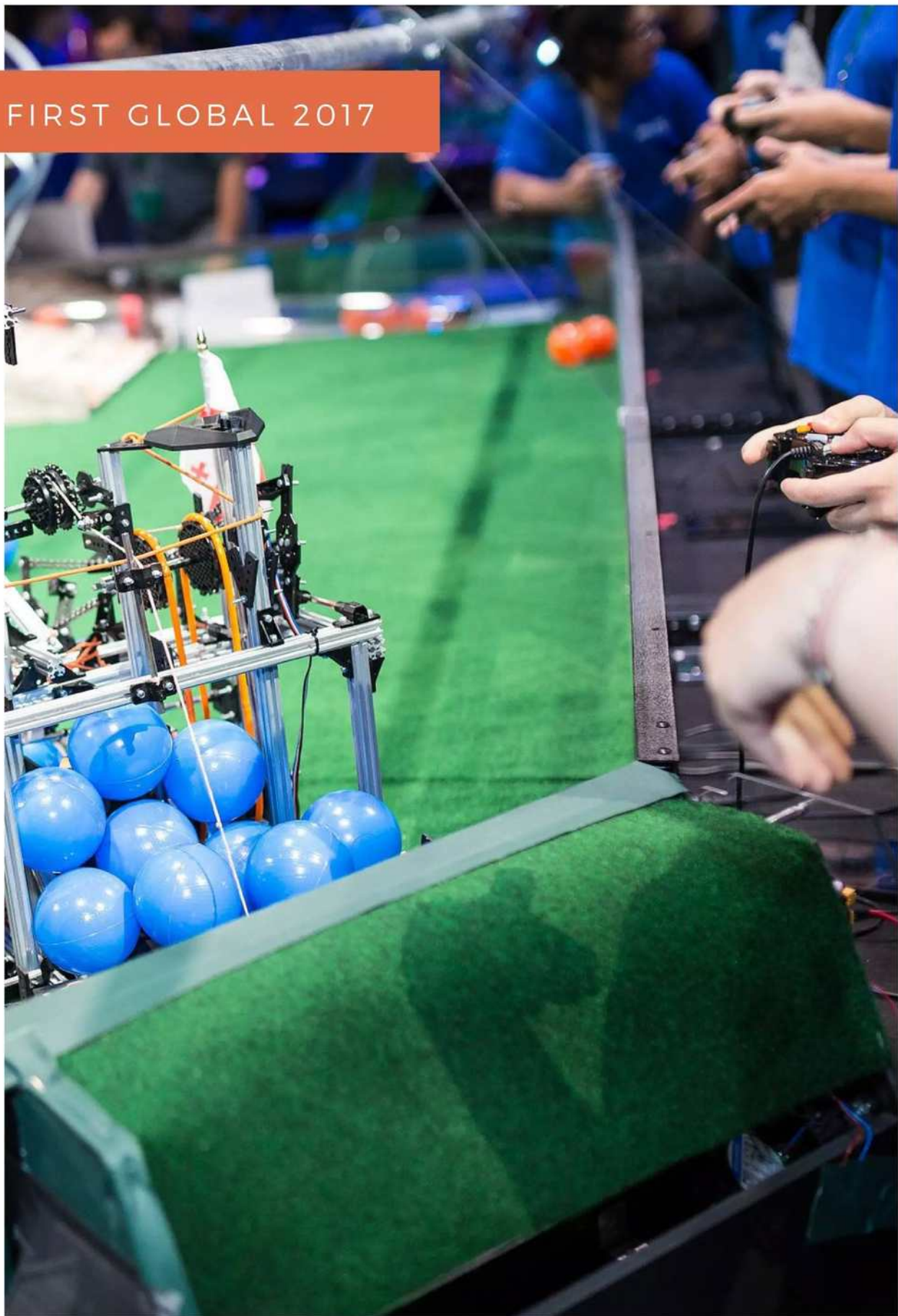


CONS:



- It does not support any other controller except their own REV controller, and that goes the same for the sensors, motors.
- There is just one competition that supports REV so far: FRC, FTC and First Global only.
- They do not have a big community and support around the world like LEGO Mindstorm or TETRIX.
- One set or kit is not enough for the student to join one of the competitions, most of robots tasks require more parts and motors that should be bought and obtained separately.
- There are not many options for coding and programing REV.
- It's only available in USA, and it's so hard to ship or get sets and extra extensions, especially for Syria because of economic sanctions on the country of US products.

FIRST GLOBAL 2017



6- ARDUINO

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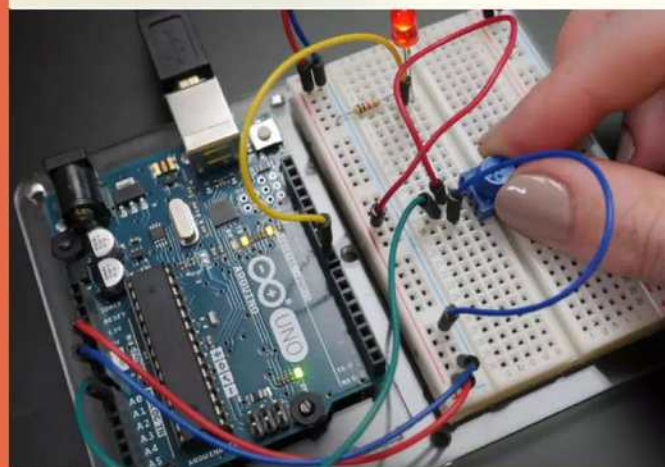
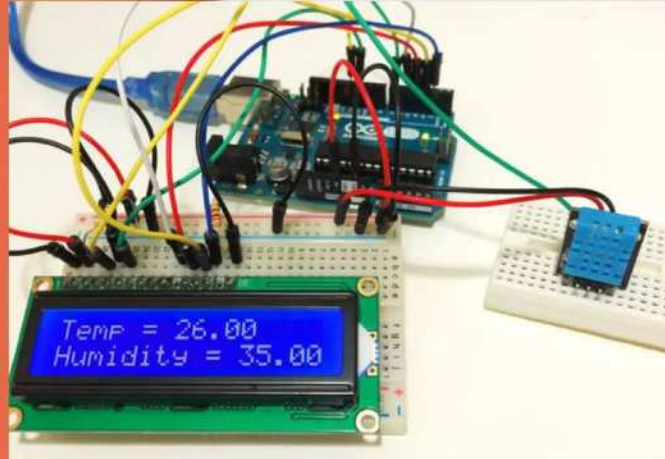
Arduino is an open-source electronics platform based on easy-to-use hardware and software. It's intended for anyone making interactive projects. Arduino senses the environment by receiving inputs from many sensors, and affects its surroundings by controlling lights, motors, and other actuators. You can tell your Arduino what to do by writing code in the Arduino programming language and using the Arduino development environment.

Arduino is considered as the main controller that can work almost with any sensors, motors and even other controllers like LEGO EV3 or NXT. Thanks for open source license that Arduino supports you can have a access to a lot of "Do it yourself" (DIY) articles and resources, access to unlimited sensors with different types and the same goes for the motors.

Arduous is considered the ultimate tools to do anything you want in the Robotics world, and the world of electronics and technology in general. And it's the cheapest among all Robotics tools available in the market.

PROS:

- It has the cheapest controller, tools in comparison with other robotics tools.
- It can work with various types of sensors, motors and other controllers.
- They have a large community and resources to find help, inspiration and more.
- It's available to buy from any place and there are no limits or rules to get and buy in the Syrian market, even there are some version of Arduino made totally inside Syria.
- Students here work with really tools with really electronics tools, not like other Robotics tools which are considered at the end of the day as toys for kids.



CONS:



- It is only supported in 2 global competitions: (WRO for open category only) and IRC (International Robotics Challenge) only. With support for one regional competition ARC (Annual Robotic Competition).
- It's quick to damage, making its lifespan short.
- You can not find any parts or tools to build the robot body, you must build this body on your own using metal, cartoon, wood or even plexi.
- It is difficult to teach for young children and its programming, because it deals with the laws and connections of electronics and requires experience in coding.
- There are so many non-original, fake parts of motors and sensors in the market and it is difficult to obtain motors and sensors of high quality at reasonable prices, and without the need to request and buy it online outside the country.

ARC SYRIA - 2017



ADDITIONAL USEFUL TOOLS

This is a list of additional tools that we can add for our robots to get more accurate sensitivity for sensors, or better handling of the software algorithms needed by some competitions, and for some age-specific groups. And you can find on this list some kits that are not suitable for competitions, but are considered suitable for the robotics labs in schools or other robotics clubs:

HiTechnic:

HiTechnic manufactures a range of robotic sensors for the LEGO MINDSTORMS NXT, EV3, TETRIX and Matrix.

These sensors have more accurate sensitivity in comparison with LEGO sensors and it adds new types, options for sensors to use. The company is also responsible for making the special ball for WRO Football competition.



KNRm Robotics:

It's another high end controller that can handle a lot of processing for big Algorithms like image processing, more motors and sensors with full support for LEGO MINDSTORMS NXT, EV3, Matrix only.

It can be programmed with NXT-G, LabView and Robot C. this controller is targeted in the advanced category and open category in WRO competition.



myRio:

It's a high end controller and it's made by National Instruments, a company who owns LabView program and many other hardware and software solution in automation field all around the world. The main purpose of this controller is for educational activities as the company say, but it can only be used with TETRIX tools, that support advanced category WRO. It is difficult to buy for Syria.



MINDSENSORS:

Mindsensors provides high quality parts for Raspberry Pi, LEGO Mindstorms NXT and EV3 Robotics systems. They have some really useful tools that can get LEGO Mindstorms NXT and EV3 to next level, such as: NXT Camera shield so you can make image processing using EV3 directly, or you can add Servo shield so you can control servo motor from your EV3, and another shield to connect Arduino with your LEGO motor and sensors and more from other shields for different tasks and different sensors. It can be used for WRO Open category and for school activities.



Raspberry Pi:

The Raspberry Pi is a series of small single-board computers, and it's open source board, like Arduino, it has a large community and resources around the world. The main purpose of these small computers is for educational activities, and it can be used in various technology projects and one of these projects are robotics projects who depend totally on help with other tools to build complete robots. Raspberry Pi is famous thanks for its cheap price and for its great coding experience, because you can install a whole operating system on it like Linux and Windows and use the full power of the designated OS to make your robot work and even connect to the internet. For Robotics competitions it can only be used in WRO open category.



LittleBits:

littleBits is a kit for kids aged 5 to 10 this kit makes technology more fun, easy-to-use, and infinitely creative for this age range. The kits are composed of electronic building blocks that are color-coded, magnetic, and make complex technology simple and fun. Using this kit can be interchangeable in millions of different ways to empower kids to invent anything - from a sibling alarm, to a wireless robot, to a digital instrument.

There are no competitions that support this kit yet, but it's considered one of the most easy and fun way to start in robotics and technology activities like coding in robotics labs, schools or other robotics clubs.



FABLAP

*We can not talk about robotics tools without mentioning FabLab.
It's kind of a robotic lab but with bigger goals and missions than any normal lab.
It adopts the concept of learning, innovation and sharing.*



CORE EQUIPMENT FOR FABLAB:

- 1- 3D PRINTER OF PLASTIC
- 2- CNC MACHINES
- 3- PRINTED CIRCUIT BOARD
MILLING OR ETCHING
- 4- MICROPROCESSOR AND
DIGITAL ELECTRONICS DESIGN,
ASSEMBLY, AND TEST
STATIONS
- 5- CUTTERS, FOR SHEET
MATERIAL

"Cities have the capability of providing something for everybody, only because, and only when, they are created by everybody."

-Jane Jacobs

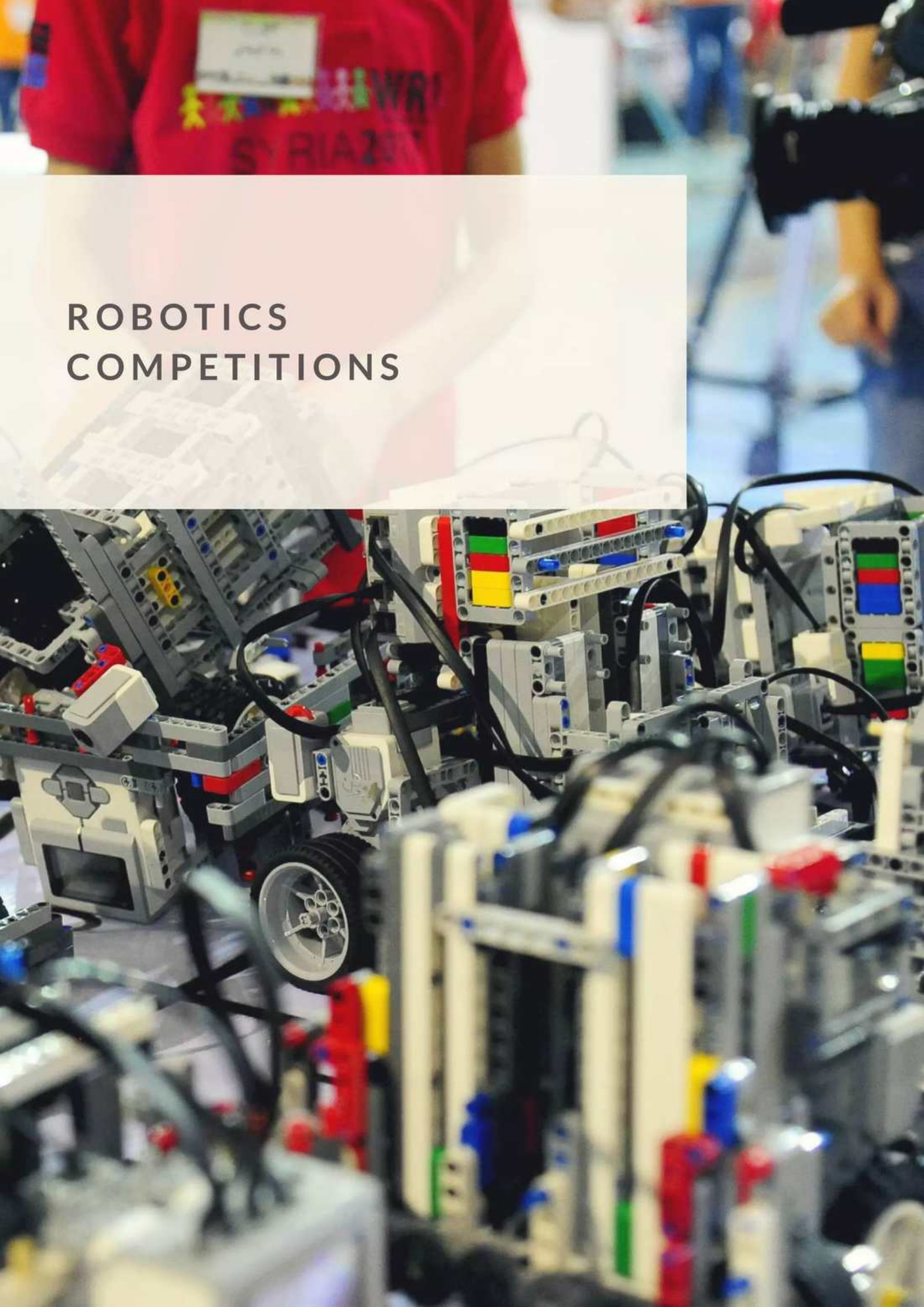
A Fab Lab is a technical prototyping platform for innovation and invention, providing stimulus for local entrepreneurship. A Fab Lab is also a platform for learning and innovation: a place to play, to create, to learn, to mentor, to invent. To be a Fab Lab means connecting to a global community of learners, educators, technologists, researchers, makers, entrepreneurs and innovators. A knowledge sharing network that spans 30 countries and 24 time zones. Because all Fab Labs share common tools and processes, the program is building a global network, a distributed laboratory for research and invention.

Having a Fab Lab in every school is essential, a much larger activity than just a robot lab, it's a community, a place that will provide the tools, assistance and information necessary to build anything related to technology in all its functionality without exception.

The fab lab movement is closely aligned with the DIY, the open source hardware and the free and open source movement, and shares philosophy as well as technology with them.



FABLAB RUC, ROSKILDE UNIVERSITY - DENMARK

The background of the page is a composite image. The top portion shows a person wearing a red t-shirt with a colorful logo and the letters 'WR' and 'S' visible. The bottom portion shows a close-up, detailed view of a complex LEGO Technic robot. The robot is constructed from various grey, white, and colored (red, blue, yellow, green) Technic beams and connectors. It features a black rubber tire on a white hub, several black cables, and a motor. The robot's structure is intricate, with many small components visible. The text 'ROBOTICS COMPETITIONS' is overlaid on the top portion of the image.

ROBOTICS COMPETITIONS



WRO

Word Robot Olympiad



Starting in 2004, World Robot Olympiad has developed to a truly global competition. WRO tournaments are organized in more than 60 countries and each season new countries join the movement. The WRO international finals in November is hosted by a different country each year.

Teams can participate in four different categories with different age groups. This allows the participants to develop their robotic skills from a very young age up to being a college or university student.

A team consists of two or three students and a coach. Participating teams need to design, build and program a robot model that is capable to perform a challenge, play soccer or demonstrate a solution for a real-life problem.

Regular Category, Open Category. These two categories are divided into 3 age groups:

Elementary: younger 12

Junior: 13-15

Senior: 16-19.

There are Football Category from age 10-19 as one group age. And there is an Advanced Robotics Challenge for age 17 - 25.

And in WRO 2017 they launched two Pilot projects: Open Category - Future Innovators so it will be another round of judging that focus on business side of the project.

WeDo Pilot to add and support WeDo kit for WRO, so small ages 5-9 can join the competition using only LEGO WeDo kit.

Every year the host country decides the theme for the competition.

Regular Category teams have a table about 2*1 meter of size, with one robot in 2 minutes only the robot must do some task on this table. There are no extra points the team can get, just one chance and what your robot can do in those 2 minutes.

Open Category teams they do not have any limitation just in the idea and place that shows the project and it's 2*2 meter booth. There are some extra points for the ideas, programming, engineering, and teamwork. But it's famous for its huge cost of money, time, and work for this kind of category.

Football Category: there are two robots they compete against two other robots they are known as Guard and attacker. This kind of competition is so new in Syria and requires high skills in programming and engineering.

Advanced Robotics Challenge: and that competition is not so famous inside Syria because of its huge cost for tools that teams must get it to start working in this competition. And the students who join this competition must have good experience in Robotics competitions and how point are get and programming and engineering skills too.

Syria hosted 5 WRO event as national event, while participating in 4 global competitions in UAE, Russia, India and finally on 2017 in Costa Rica



Is an international youth organization that operates the FIRST Robotics Competition, FIRST LEGO League, FIRST LEGO League Jr., FIRST Tech Challenge and FIRST Global competitions.

FIRST seeks to promote a philosophy of teamwork and collaboration among engineers and encourages competing teams to remain friendly, helping each other out when necessary. And supports respect towards one's competitors and integrity in one's actions.

FIRST LEGO League Jr.: ages 6-10
 FIRST LEGO League: ages 9-14
 FIRST Tech Challenge: ages 12-18
 FIRST Robotics Competition: ages 14-18
 FIRST Global: ages 15-18

The beginning of robotics in Syria started with FIRST LEGO League FLL, but after 2012 the FIRST stopped its support to buy or join regional, global competitions because of US sanctions on Syria that does not allow any US company, organization in tech field to interact with those countries.

Every year there is one theme for every competition in FIRST, but the difference between FIRST and WRO is that FIRST does not focus 100% on the robot's work, engineering and programming, there are a lot of factors that teams can get their points from their projects, every team must provide a solution and very invitation one as projects and present this project for panel of judging, so the team can get almost 30% of points for the project and 35% from the robot and 35% from the core value, and how they are active and interact with other participants.

"FIRST is more than robots. the robots are a vehicle for students to learn important life skills. Kids often come in not knowing what to expect- of the program nor of themselves. they leave, even after the first season, with a vision, with confidence, and with a sense that they can create their own future" the founder of FIRST Dean Kamen





FIRST GLOBAL



From the same organization FIRST (For Inspiration and Recognition of Science and Technology) they started on 2017 the first FIRST Global Olympics-style robotics very similar to WRO, but with FIRST spirit in it.

It was founded by the co-founder of FIRST, Dean Kamen, with the objective of promoting STEM education and careers in the developing world through Olympics-style robotics competitions.

The 2017 FIRST Global Challenge was held in Washington, D.C., from July 17–19, and the challenge was the use of robots to separate different colored balls, representing clean water and impurities in water, symbolizing the Engineering Grand Challenge (based on the Millennium Development Goal) of improving access to clean water in the developing world. Around 160 teams composed of 15–18-year-olds from 157 countries participated, and around 60% of teams were created or led by young women. Six continental teams also participated.

Mexico City would host the 2018 FIRST Global Challenge.

Syria was supposed to participate with one team but VISA limits stopped Syria team to join in FIRST Global 2017. Note that teams for Afghanistan, Sudan, and Iran got the VISA, but some of these teams could not get their own robotics kits, so other teams in the US worked with these teams by collaborating online to build their robot remotely, as a kind of cooperation and communication with the participants and helping each others.





Techfest
2017-18

IRC

International Robotics
Challenge



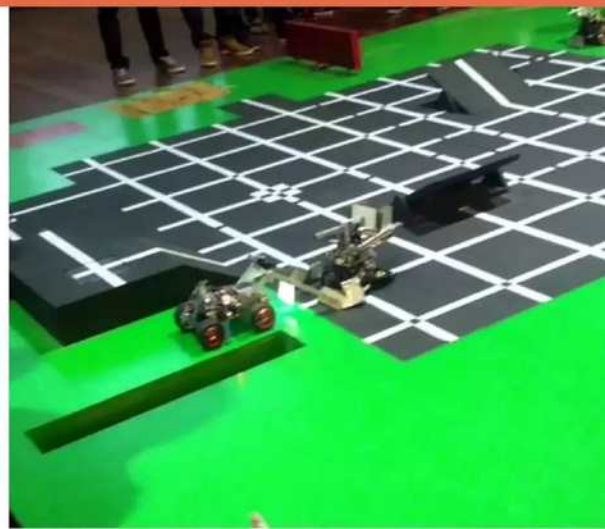
IRC competition is a part of Techfest which is the annual science and technology festival of Indian Institute of Technology Bombay. This festival has a lot of activities like: Exhibitions, workshops, lectures, summits and more. IRC one of many competitions that happen in Techfest; there are about 12 countries that join the IRC competitions every year in Mumbai, India.

Maximum team members are 5, with no limitations on the age expected to by a student from any college or university.

Syria will join the IRC for the first time in 2017 with one team.

IRC task:

1. There will be two bots, a manual bot and an autonomous bot. They need to coordinate with each other to complete the task.
2. The autonomous bot has to solve the grid by following the white lines while avoiding the node, with help of image processing transfer blue block in a transfer zone which would be based on the direction of arrow on the sign board, put red block in red pit, cross the incline and Seesaw, transfer block with Techfest logo to third transfer zone.
3. The manual bot has to cross curved and regular bridge, transfer the blocks from transfer zone to pits and throw magnetic dart on the target.
4. A maximum of 7 minutes will be allotted to each participating team.





It's a regional competition happening in Syria by SCS (Syria Computer Society) and Lebanon by Kalimat organization. This competition which targets ages of 10-25 years and focus on Arduino Tools for their competition tasks, because this kind of tools is easy to get for schools and other robotics clubs, with the intention of spreading the culture of Robotics in the region.

Syria organize local competitions and the winners of these competitions go and join Lebanon competitions.

ARC 3 in Lebanon hosted more than 100 teams from different 60 schools an robotics clubs from Lebanon and Syria.

ARC have many categories and they are:

Mini Maze: for primary students grades 1, 2 and 3, the task is that the robot must find its way out of the maze

Football Robot Game: for primary students grades 4, 5 and 6 and there are two robots facing another two robots in a special football field.

Robot Race: For middle school students, and their task is to build a robot to follow a line on a different map.

Sumo Robot: for high school students and university level, two robots face each other and the winner is the one who pushes the other outside the ring.

Technology Project: For high school students and university level, it is similar to the WRO Open Category, each year there is a theme and the students must build a solution for this year's theme.



OTHER COMPETITIONS WORTH MENTIONING:

Sure there are lots of other competitions around the world. And here is a short list for some of these competitions: Remember this is just a short list of great other competitions or experiences that are available all around the world that could be applied in our community:

EDC:

Engineering Design Challenge (EDC) is a regional university-level engineering competition. It was launched for the first time in the American University of Beirut by the AUB Robotics Club in 2015. There was a local competition to join the EDC 2.0 2017 in Syria organized by Kazen team.



OARC:

Open Arab Robotic championship, it is a regional competition with almost all Arab countries, each year the competition happens in one of the Arab countries. There are different categories for this championship and they are: FLL, FIRST Jr. FTC, Line Tracking Robot, Robot Sumo, Ball collector, and Open Arab Challenge. And there are almost 100 teams competing yearly.



VEX Robotics Competitions:

is an International competition that happens every year and depends on the FIRST FTC model so there's a local competition and the winner from every country can join the worldwide competition. Participants in these competitions can only use VEX kits and materials. And there are about 30 countries that compete in VEX Robotics World Championship yearly.



MBZIRC:

Mohamed Bin Zayed International Robotics Challenge, it's a global competition in Abu Dhabi, UAE occurs every two years. The competition accepts students from university level and up. The special thing about this competition is its use of large open fields and interaction with other objects, for example, one of the challenges in 2017 was landing a drone on a moving vehicle and the competition's prize is about 5 million U.S. Dollar.



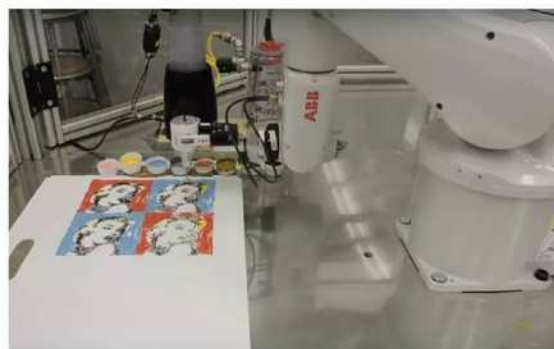
ITU Robot Olympics

Istanbul Technical University Robotics Olympics, is a local competition happening in Turkey they have various categories and they encourage all the students the main attraction for this local competition is the focus on collaboration in their robotic club and the spirit of the technical and robot idea in the community.



RobotArt:

The \$100,000 Robot Art competition, The contest's goal is to challenge teams to produce something visually beautiful with robotics – that is, to have a robot use physical brushes and paint to create an artwork. Many teams from around the world join this competition and there's a lot of great artworks, and that encourages the collaboration between technical fields and art fields.



RobotGeams:

Is the Olympics of Robots - we invite the best minds from around the world to compete in over 50 different events: combat robots, fire-fighters, LEGO bots, hockey bots, walking humanoids, soccer bots, sumo bots, and even androids that do kung-fu. Some robots are autonomous, some are remote controlled - but they're all cool! As an open event, anyone can compete.





ROBOTICS CLUB
IN SYRIA

ROBOTICS CLUB

IN SYIRA



To make and establish a robotics Club in school or in any other Institution, organization you have to meet 4 important component for this club and they are: The coaches, your robotics tools and materials, money budget, and the students.

Know that establishing a robotic club is something more important for your community and students more than just making this robotics club to join the local, and global competitions.

THE COACH

Let's start with number one, and it is the coach: A coach or a mentor or a teacher is the most important person in any robotics club, how much skills the coach has reflects directly to how much the team skills and level will be on it.

The main tasks for the coach is to manage and train and teach and mentor the team to achieve one goal and it's to teach kids robotics and that means programming, engineering, problem solving skills, creativity, teamwork and all our experience in life like time management, and how to work in a team and more other skills the robotics experience can provide for them.

The second goal of coach should be joining the competition to win the first place and to learn as much as they can.

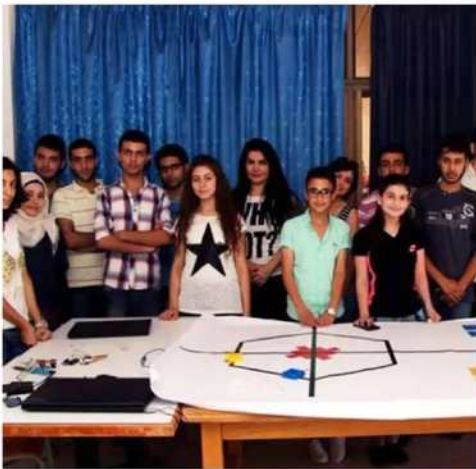
A good coach can make the training hour like more than 5 months of fun time for the students, however, a bad coach can really make this time very horrible for students and give them a bad experience about robotics and rally waste their time.



ROBOTICS CLUB



Why the coaches are Interested in this kind of job? I really ask a lot of questions about why they are interested in this kind of job admission and there are a lot of responses I can shortlist it for you: The first reason for most coaches that are interested in training kids is to build up their CV and resumes most of them see Robotics as a trend mark and at the same time they are old enough to be a coaches, and they don't have that chance to be competitors since since most competitions' age cap is 18, and at the same time their universities do not have the equipment necessary for an advanced robotics challenge for university students, so most of them are interested in this field and the only opportunity to go on this field is to be a coach.



second point is that most of the coaches are university students so some of them see the robotic coach is like a part-time job that provides for them some money and at the same time it don't affect negatively on their study time.

And other coaches see there's that opportunity if they work well with the team to travel outside the country, and live an experience that's happening with the students. They don't really care about money in the first place and they are looking for the end results of the team.

After the coaches finish their studies in university and they get enough experience from previous seasons of the robotics competitions or of training kids, sadly the coach can not remain or continue In this kind of job because their responsibilities is growing when they are graduating from the university and there is a need to get a full time job with a really good salary unlike the part time job with low financial compensation. Or maybe that coach got enough experience to use in their resume, or even the coach got what they wanted by traveling outside the country and joining a Global competition.

And here the lifespan of the coaches are in danger because the coach gets enough experience and a great one that really can help make the robotics Clubs and their students get to another level, but because of lack of money, and because they lack office experience to use in their CV or maybe the coach have their personal goals. Because of that the lifespan of coaches in a robotic club is going as 3 years on average,

And in order extend the lifespan of coaches in the robotics clubs we must work and provide solution for the coaches, like extra money, and put in their mind that the good skilled coach could be a full time employee inside the robotics clubs of course, raising the money resource for robotics clubs and budget could contribute to extend their lifespan, or getting an official experience certificate from university or another organization or government institution

ROBOTICS CLUB

ROBOTIC EQUIPMENT AND MONEY:

The second component for robotic club is the robotic equipment, tools, parts and resources themselves. the size of robotic club and how much students they got there and the level of the team in the competition directly depends on how much equipment they have in the robotic Club.

And this component is connected directly with the third component of any robotics club and that is the money, how much the robotic club could get money and revenue and sponsorship enough to finance the robotic club.

Enough money for rugby club that means enough robotic equipment, tool. The money role here is more than buying robotic equipment, the money is also responsible to fund the travel expenses for local and global competitions, the entry fee, the coaches salaries, the rented place, and the growth of robotics club, with some marketing for the club.

And here the robotic club must be creative in how to get income and funds. For example, they could get income from organizing a workshop, event, technical festival making courses for students and get them pay fees for entry, starting advanced courses, TOT (Train of trainer) courses, getting sponsorship like adding the sponsor's company logo on their t-shirts at a local or global competition and in their marketing channel, try to connect and get funding from global organizations like UNICEF or UNDP, UNFPA and more.

The administration and the manager of robotics club they must spend a lot of time, effort, connections and networking to get enough money for the club every year.



ROBOTICS CLUB

THE STUDENT:

The last component on robotic club is the student and they are the most important and essential part the student will join the robotic Club as a beginner and your mission as a coach or as manager it to give and provide Not forgetting a great experience of teaching coding programming creativity Inspiration problem solving teamwork management of time researcher and search and more of other experience that's can get it from your robotic courses.

not all student of course like this kind of Films are science and maybe their parents force them to join This field and your missions to to guide them for the right field of them maybe some of student love art love sport or music and your job to mentor them and their parents to the right place

Not all student have equal level some of them are great on math and programming or engineering so they are fit directly to a team that represent the robotic club on the local or global competition and this kind of student they need your attention to get them level up and give him a lesson that the winner of the competition is not everything.

Some of the students are not in the required level to join the competition and you should give him more and more attention for them to get them up level and training them and preparing them for next year competition, And you really need to focus on joining competition is not everything and focusing in the most important thing on the skills and the knowledge that the student get it from this experience and that's what's making robotic as good as a unique experience for everyone Parents coaches student and the whole community.





KIDS WHO DREAM
BIG



Leen Story

At the end I would like to share a story with you my little sister Leen:

This is Leen. Leen is nine years old, and her favorite subject is math.

In two weeks her brother will invite her to join her first robotic team.

Two months after that she'll use basic coding to bring her idea to life, then in a year she'll meet her best friend at a regional robotics tournament.

Two years after that she'll lead a team that wins third place in a global innovation challenge, when Leen graduates college she'll intern for an engineering firm improving water service solutions in developing countries she'll see the world.

One year later she'll land her first job as an electrical systems engineer six years after that showing a genius award for her work and renewable energy.

20 years later she'll show her daughter how to follow in her footsteps.

Robotics support kids who dream big your action makes those dreams a reality.

ABOUT THE AUTHOR

TAREK SHEIKH AL-SHBABA



22 years, study computer engineering at technological computer college, Damascus University with double major Management information system at faculty of Business Administration, Arab International University.

Passionate about Entrepreneurship and Startup companies, and I'm love this kind of environment that everyone in the team working for the same goal, and hustling to achieve it.

I work with youth and kids in last 4 years with robotic and programming field and I teach them robotic as programming, mechanisms how to build the robotic from motor, sensor and code, and for that I build my own tools and curriculum.

In addition for that, I work with this youth for their open project in tech and they make their research, how to manage the time, how to present their idea and how they sell their idea.

